

TUTORIAL 4(14/8/26)

1. An online shopping store wants to develop a Product Billing System to calculate the final price of products after applying discounts. Create a Java program using a **Product class** and **objects** to store the following details of 5 products (Use **Array of object**, **Looping** and **conditional control structures**)

- Product ID
- Product Name
- Price
- Quantity

Create an **array of 5 Product objects** to store the product details.

calculate the **total price** of each product. If the total price is **₹5,000 or more**, give a **10% discount**. Otherwise, give a **5% discount**. Calculate the final price using:

Finally, display the **Product ID, Product Name, Total Price, Discount, and Final Price** for all 5 products.

Sample Input and Output:

Product 1:

ID: 101

Name: Laptop

Price: 50000

Quantity: 1

Output:

Product ID: 101

Name: Laptop

Total Price: ₹50000

Discount: ₹5000

Final Price: ₹45000

2. A college wants to develop a Student Performance Management System to evaluate the academic performance of its students. (Constraint: Choose an appropriate decision-making operator/construct for the conditions . Avoid using multiple lines of branching code)

Create a Java program using a Student class with the following attributes:

- Roll number
- Student name
- Marks in 3 subjects
- Attendance percentage

Create 5 Student objects and store them in an array of Student objects.

For each student:

1. Calculate the total marks and average marks using a for loop.
2. Determine whether the student has Passed or Failed based on the average:
 - $\text{Average} \geq 50 \rightarrow \text{Pass}$
 - $\text{Average} < 50 \rightarrow \text{Fail}$
3. Determine whether the student is eligible for a scholarship:
 - $\text{Average} \geq 75$ and $\text{attendance} \geq 80\% \rightarrow \text{Eligible}$
 - Otherwise $\rightarrow \text{Not Eligible}$
4. Classify the student's performance:
 - $\text{Average} \geq 85 \rightarrow \text{Excellent}$
 - Otherwise $\rightarrow \text{Good}$
5. Display the student's details along with the above results.
6. Finally, identify and display the student with the highest average.

3. A company wants to develop a simple Employee Management System to calculate the yearly salary and bonus of its employees. The HR department needs to store the details of 5 employees and determine which employees are eligible for a bonus.

Develop a Java program that:

- Creates an Employee class containing employee details such as Employee ID, Name, and Monthly Salary.
- Stores the details of 5 employees using an array of objects.
- Creates methods to:

- Display the details of each employee.
- Calculate and display the annual salary.
- Calculate the bonus based on the employee's salary.
- Check and display whether the employee is eligible for a bonus.
- Use a suitable condition to determine bonus eligibility. For example, an employee earning a monthly salary of ₹30,000 or more is eligible for a 10% annual salary bonus.
- Display the complete details of all employees along with their annual salary, bonus, and eligibility status.

4. A multispecialty hospital wants to automate its consultation billing process. The billing department needs to maintain the consultation details of 5 patients and calculate the final amount payable after applying a discount. Develop a Java program for this system.

- Create a Patient class with patient name and consultation fee.
- Use a parameterized constructor to initialize the patient details.
- Store the details of 5 patients using an array of objects.
- Create a method with arguments to calculate the final consultation amount.
- Apply discounts
 - Consultation fee $\geq$ ₹2000 $\rightarrow$ 10% discount
 - Consultation fee $<$ ₹2000 $\rightarrow$ 5% discount
- Use a loop to process all 5 patients.
- Create a separate method to display:
 - Patient name
 - Original consultation fee
 - Discount
 - Final amount
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5. An amusement park wants to display the activity level of its rides in a simple visual format. The park has 5 different rides, and each ride is assigned an activity level from 1 to 5 based on the number of visitors.

Develop a Java program to generate the following pattern:

Ride 1: *

Ride 2: **

Ride 3: ***

Ride 4: ****

Ride 5: *****

- Create a Ride class with ride number and ride name.
- Use a constructor to initialize the ride details.
- Store the 5 rides using an array of objects.
- Use a for loop to access each ride.
- Use a nested for loop to print the * pattern.
- The number of * should correspond to the ride number/activity level.
- Display the ride name along with its corresponding pattern.

Sample Output

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Roller Coaster : *
Ferris Wheel   : **
Bumper Cars    : ***
Water Ride     : ****
Haunted House  : *****
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